ENGINEERING DRAWING WITH AUTOCAD

UNIT 1: INTRODUCTION TO ENGINEERING DRAWING & AUTOCAD INTERFACE

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Engineering Drawing with an Introduction to AutoCAD by Dhananjay Jolhe

>> 1.1 Conceptual Framework of Engineering Drawing

Engineering drawing is the universal graphic language of engineers and technicians used to communicate technical ideas, geometric specifications, and manufacturing requirements without ambiguity:
* Universal Standards: Governed internationally by ISO (International Organization for Standardization) and in India by BIS (Bureau of Indian Standards - SP 46:2003).
* Drawing Instruments & Equipment: Drawing board, T-square or Mini-drafter, Set-squares (45-45 and 30-60), Engineering scale, Clinograph, Compasses (large and spring bow), Dividers, French curves, Drawing paper sheets.
* Standard Sheet Sizes (ISO A-Series based on golden ratio 1:sqrt(2) with area of A0 = 1 sq.m):
- A0: 841 mm x 1189 mm (Area = 1 m^2)
- A1: 594 mm x 841 mm
- A2: 420 mm x 594 mm
- A3: 297 mm x 420 mm (Standard engineering lab drawing sheet)
- A4: 210 mm x 297 mm (Standard project report & office documentation)
* Title Block: Positioned at bottom-right corner of sheet (recommended size 185 mm x 65 mm as per BIS). Contains: Name of Institution, Project/Assignment Title, Scale, Projection Symbol (1st/3rd Angle), Drawing Number, Date, and Approver Signatures.

>> 1.2 Line Types and Graphic Conventions (BIS SP 46)

Engineering lines carry specific semantic meaning based on thickness and pattern:
* 1. Continuous Thick Line (Grade H/HB, ~0.5 mm): Visible outlines, visible edges, and crests of screw threads.
* 2. Continuous Thin Line (Grade 2H/3H, ~0.25 mm): Dimension lines, extension lines, leader lines, projection lines, construction lines, and hatching/section lines.
* 3. Continuous Thin Freehand Line: Boundaries of partial or interrupted views and sections.
* 4. Dashed Thin Line (Grade 2H, ~0.25 mm): Hidden outlines and hidden edges (dash

length 3 mm, gap 1 mm).
* 5. Long-Dashed Dotted Thin / Chain Thin Line: Center lines, lines of symmetry, pitch circle diameters of gears, and trajectory paths.
* 6. Chain Thin Line with Thick Ends: Cutting plane lines indicating where a cross-section is taken.

>> 1.3 Dimensioning Principles and Rules

Dimensioning is the art of providing precise numerical values of geometric sizes and locations:
* Elements of Dimensioning: Extension line (projects 2-3 mm past dimension line), Dimension line (parallel to measured feature), Arrowheads (proportions 3:1 length to width, fully filled), Leader line (thin line inclined at 30-60 deg ending in horizontal landing of 2-3 mm).
* Dimensioning Systems:
- 1. Aligned System: Dimensions placed parallel to and above dimension lines. Read from bottom or right edge of sheet.
- 2. Unidirectional System: All dimensions placed horizontally, breaking the dimension line near center. Read exclusively from the bottom.
* Golden Rules: Never cross dimension lines with other lines; dimension outer features outside the view; avoid redundant dimensioning; dimension hidden features only when unavoidable.

>> 1.4 Engineering Lettering: Single Stroke Vertical Gothic

* Standard: Lettering must be plain, uniform, and legible as per BIS SP 46.
* Single Stroke: The thickness of the line is determined by a single pencil stroke (continuous line thickness).
* Vertical Gothic: Upright capital letters perpendicular to the baseline.
Height-to-width ratio typically 7:5 (except W=7:7, I=7:1).
* Standard Heights: 2.5 mm, 3.5 mm, 5.0 mm, 7.0 mm, 10.0 mm, 14.0 mm, 20.0 mm. Titles use 7-10 mm; subtitles use 5 mm; dimensions & notes use 3.5 mm.

>> 1.5 Theory of Engineering Scales

Scale is the ratio of linear dimensions of an object in a drawing to actual dimensions:
* Representative Fraction (RF): RF = (Length of object on drawing) / (Actual physical length of object). Units in numerator and denominator must match.
* Types of Scales:

- Full Scale (RF = 1:1), Reducing Scale (RF < 1, e.g., 1:2, 1:5, 1:100), Enlarging Scale (RF > 1, e.g., 2:1, 5:1, 10:1).
* Maximum Length to Measure (ML) & Length of Scale (LOS): LOS = RF x ML.
* Plain Scales: Measures two consecutive units (e.g., meters and decimeters, or centimeters and millimeters).
* Diagonal Scales: Measures three consecutive units or fractions down to two decimal places (e.g., meters, decimeters, centimeters) using the geometric principle of similar right triangles.

>> 1.6 Introduction to AutoCAD Interface and Environment

AutoCAD is the industry-standard Computer-Aided Design software developed by Autodesk:
* Interface Elements: Application Menu, Quick Access Toolbar, Ribbon Tabs & Panels, Drawing Area (Graphic Window), Command Line Prompt, Status Bar, UCS Icon, ViewCube.
* Key Setup Commands:
- UNITS: Sets unit type (Decimal, Architectural) and precision (e.g., 0.00).
- LIMITS: Defines the electronic drawing boundary (Lower left: 0,0; Upper right: 420,297 for A3).
- ZOOM -> ALL (Z -> Enter -> A -> Enter): Resets display to encompass established drawing limits.
* Essential Drafting Aids (Function Keys):
- F1: Help system | F2: Expanded text command window | F3: OSNAP toggle (Object Snap: Endpoint, Midpoint, Center, Intersection).
- F7: Grid display toggle | F8: ORTHO mode toggle (restricts cursor to 0, 90, 180, 270 deg).
- F9: SNAP toggle (locks cursor to grid increments) | F10: Polar Tracking | F11: Object Snap Tracking.
- F12: Dynamic Input toggle (displays command prompts and coordinates directly near cursor).
* Coordinate Systems: Absolute Cartesian (X, Y), Relative Cartesian (@dX, dY), Relative Polar (@dist<angle), and User Coordinate System (UCS).

[PROBLEM 1: Construction of a Plain Scale for Architectural Measurement]

Problem Statement: Construct a plain scale of RF = 1:50 to show meters and decimeters, capable of measuring up to a maximum distance of 6 meters. Mark a distance of 4.7 meters on this scale.
Mathematical Derivation & Construction Steps:
* Step 1: Calculate Length of Scale (LOS):

- LOS = RF x ML = (1 / 50) x 6 m = (1 / 50) x 600 cm = 12.0 cm.
* Step 2: Main Scale Division:
- Draw a horizontal rectangle of length 12 cm and width 1 cm.
- Divide the 12 cm length into 6 equal parts using acute angle line division method.
Each part represents 1 meter (12 cm / 6 = 2 cm per meter).
- Number the main divisions: Label the first division mark from left as 0, then 1, 2, 3,
4, 5 to the right.
* Step 3: Sub-Division of First Main Division:
- The leftmost 1-meter division (from 0 to the left) is divided into 10 equal parts.
Each sub-part represents 1 decimeter (0.1 m).
- Label sub-divisions to the left: 0, 2, 4, 6, 8, 10 decimeters.
* Step 4: Marking the Target Distance (4.7 meters):
- From the 0 mark, extend a measurement 4 units to the right (4 meters).
- From the 0 mark, extend 7 units to the left (7 decimeters = 0.7 meters).
- Total measured distance = 4 m + 0.7 m = 4.7 m. Draw extension lines and dimension line
with arrows and label 4.7 m.

[PROBLEM 2: Design and Construction of a Diagonal Scale]

Problem Statement: On a railway map, an actual distance of 36 km is represented by a 9
cm line. Construct a diagonal scale to read kilometers, hectometers, and decameters, up
to a maximum distance of 60 km. Mark a distance of 43.6 km (or 43 km and 6 hm).
Mathematical Derivation & Construction Steps:
* Step 1: Calculate Representative Fraction (RF):
- RF = Drawing length / Actual length = 9 cm / (36 km x 1000 m/km x 100 cm/m) = 9 /
3,600,000 = 1 / 400,000.
* Step 2: Calculate Length of Scale (LOS):
- LOS = RF x ML = (1 / 400,000) x 60 km = (1 / 400,000) x 6,000,000 cm = 15 cm.
* Step 3: Construction of Scale Outline & Main Divisions:
- Draw a rectangle of length 15 cm and height 5 cm.
- Divide the 15 cm length into 6 equal main divisions. Each division represents 10 km
(15 cm / 6 = 2.5 cm per 10 km).
- Mark 0 at the end of the first division, and label 10, 20, 30, 40, 50 km to the right.
* Step 4: Division of Secondary Unit (Kilometers):
- Divide the first 10-km division on the left into 10 equal parts. Each part represents
1 km.
* Step 5: Diagonal Principle for Tertiary Unit (Hectometers):
- Divide the vertical height of 5 cm into 10 equal parts (numbered 0 to 10 upwards).
Each vertical division represents 1 Hectometer (1 hm = 0.1 km).
- Connect the 0 mark on top to the 1 mark on bottom, creating diagonals of slope.

* Step 6: Marking 43.6 km:
- Locate 40 km mark on the right of 0.
- Locate 3 km on the bottom horizontal sub-division left of 0.
- Move vertically up along the diagonal corresponding to 3 km to the 6th horizontal grid line (representing 6 hm = 0.6 km).
- Place arrowheads and dimension line showing exact distance = 43.6 km.

[PROBLEM 3: Standard BIS Dimensioning Conversion Case Study]

Problem Statement: A mechanical stepped bushing has outer diameter 60 mm, inner bore 30 mm, step length 40 mm, and overall length 90 mm. Convert an unaligned, chaotic sketch into a standard BIS-compliant unidirectional drawing.
Solution & Engineering Specifications:
* 1. Placement of Views: Draw Front View with centerline. Place bore hidden lines symmetric to centerline.
* 2. Linear Dimensions: Overall length 90 mm placed 15 mm below bottom edge. Step length 40 mm placed 8 mm below bottom edge (staggered to avoid crossing).
* 3. Diametral Dimensions: External diameter indicated as dia 60 (symbol circle with slash) placed outside the view. Inner bore indicated with leader line and callout: dia 30 THRU.
* 4. Arrowheads: Drawn with 3:1 aspect ratio, sharp point touching extension line, completely blackened.
* 5. Text Alignment: Numerals 90, 40, 60 placed horizontally in gaps cut into dimension lines as per unidirectional standards.

[PROBLEM 4: Complete AutoCAD Setup Workflow for Engineering Drafting]

Objective: Create a standardized engineering drawing template in AutoCAD 2024 for an A3 lab assignment.
Step-by-Step AutoCAD Command Execution Protocol:
* Command 1: UNITS -> Select Decimal, Precision 0.00, Insertion Scale: Millimeters. Click OK.
* Command 2: LIMITS -> Specify lower left corner: 0,0 -> Specify upper right corner: 420,297.
* Command 3: ZOOM -> A (Encompass entire 420x297 mm A3 workspace).
* Command 4: LAYER -> Create standard enterprise layers with color and linetype coding:
- Layer 0_Visible: Color White/Black, Linetype Continuous, Lineweight 0.50 mm.

- Layer Hidden: Color Yellow, Linetype HIDDEN2 (dash length 3 mm), Lineweight 0.25 mm.
- Layer Center: Color Red, Linetype CENTER2 (long-short-long), Lineweight 0.25 mm.
- Layer Dimensions: Color Cyan, Linetype Continuous, Lineweight 0.25 mm.
- Layer Construction: Color Green, Linetype Continuous, Lineweight 0.15 mm.

* Command 5: OSNAP -> Enable Endpoint, Midpoint, Center, Intersection, Perpendicular, Quadrant.
* Command 6: RECTANG -> 0,0 to 420,297 (Sheet border). OFFSET -> 10 mm inward for working boundary (20 mm on left margin for binding).
* Command 7: SAVEAS -> Template format (*.dwt) named LPU_CSE_Standard_A3.dwt.